SOAP API Lighweight Load Test Script

Overview

This PowerShell script is designed to perform concurrent SOAP API performance testing against the LexisNexis Bridger Insight Web Service.

The script supports:

- Multi-threaded concurrent API requests
- Connection warm-up
- Per-thread statistics
- Overall performance statistics
- Request timeout configuration
- Optional SSL certificate bypass for test environments
- RunspacePool-based concurrency optimization

Features

- Concurrent SOAP API load testing
- Thread-safe statistics collection
- Average / Min / Max response time reporting
- Success and failure counting
- Connection pooling optimization
- Configurable concurrency and request count
- Warm-up request before stress testing

Requirements

- Windows PowerShell 5.x or PowerShell 7+
- Network access to the target SOAP endpoint
- Valid Bridger Insight SOAP credentials

Configuration

Modify the following variables before running the script:

Endpoint Configuration

```
$url = 'https://<EnterpriseServerIP>/LN.WebServices/12.0/XGServices.svc/Search'
```

Replace:

```
<EnterpriseServerIP>
```

with your actual server IP or hostname.

SOAP Authentication

Update the SOAP body with valid credentials, update the SOAP body content.

```
<brid:ClientID></brid:ClientID>
<brid:Password></brid:Password>
<brid:UserID></brid:UserID>
<brid:ClientReference></brid:ClientReference>
```

Concurrency Settings

```
$ThreadCount = 2
$CallsPerThread = 10
$PerRequestTimeoutSec = 600
```

Parameter	Description
ThreadCount	Number of concurrent threads
CallsPerThread	Number of requests executed per thread
PerRequestTimeoutSec	Timeout for each request in seconds

SSL Certificate Validation

The following line disables SSL certificate validation:

```
[System.Net.ServicePointManager]::ServerCertificateValidationCallback = { $true }
```

This should only be used in testing or non-production environments.

Running the Script

Execute from PowerShell

```
.\SoapConcurrentTest.ps1
```

If execution policy blocks the script:

```
Set-ExecutionPolicy RemoteSigned
```

Output Example

```
========= Per-Thread Statistics =========
Thread 1: Planned 10, Success 10, Failure 0
  Average: 320 ms, Min: 210 ms, Max: 560 ms, Total: 3200 ms

Thread 2: Planned 10, Success 9, Failure 1
  Average: 350 ms, Min: 240 ms, Max: 610 ms, Total: 3150 ms

========= Overall Statistics =========
Thread Count: 2
Calls Per Thread: 10
Total Planned Calls: 20
Total Success Calls: 19
Total Failure Calls: 1
Overall Average Response Time: 335 ms
Overall Minimum Response Time: 210 ms
Overall Maximum Response Time: 610 ms
Overall Total Time: 6350 ms
```

Notes

- The script uses PowerShell RunspacePool for concurrency.
- Failed requests are excluded from response time calculations.
- Warm-up requests help establish persistent connections before the actual load test.
- The script is suitable for lightweight load testing.

Recommended Usage

Recommended environments:

- SIT
- Internal performance testing
- SOAP API validation testing

Not recommended for:

- Large-scale stress testing
- Production penetration testing

Author

Internal Utility Script for SOAP API Concurrent Testing